

Designing Eniac's A/B Test

Redesign of the button “SHOP NOW”



Version A: the original site.

Version B: the SHOP NOW button is now red.

Version C: the SHOP NOW button has a new text, “SEE DEALS”

Version D: both the B and the C changes to the button have been applied.

Users See Different Versions Through Random Assignment and Tracking

Core principle:

- Random assignment

Implementation:

- Tracking via cookies or user IDs
- A/B testing tools (Google Optimize, Optimizely...)

1

User visits the website

2

System checks **cookie** or **user ID**

3

User is **randomly assigned** to version A/B/C/D

4

Assigned version is stored using a **cookie** or **user ID**

5

System shows the **same version** on every return visit

6

Data is tracked

Advantages of Random Assignment in A/B Testing

- **Eliminates bias:** Users are assigned randomly, not by behavior or characteristics.
- **Ensures comparable groups:** Each version receives a similar mix of users, making results fair.
- **Isolates the effect of the change:** Differences are caused by the tested element, not external factors.
- **Improves statistical validity:** Results are reliable and testable.
- **Produces trustworthy results:** Reduces the risk of false winners and misleading insights.

3. How can we show one version to one group of users and another version to a different group?

Primary and Secondary Metrics Show Engagement and Revenue Impact of Button Changes

Primary metric

👉 **Click-Through Rate (CTR) on the button** - shows exactly how the button affects user behaviour (did users click more?)

$$\text{CTR (\%)} = \frac{\text{Total Clicks}}{\text{Total Visits}} \times 100$$

Secondary metrics - ensure clicks **translate into revenue**, not just activity

🛒 **Purchase Conversion after click** - shows how many clicks led to purchases

$$\text{Purchase Rate (\%)} = \frac{\text{Total Purchases}}{\text{Total Clicks}} \times 100$$

🎯 **Overall Conversion Rate** - measures total conversion relative to all visits

$$\text{Overall CR (\%)} = \frac{\text{Total Purchases}}{\text{Total Visits}} \times 100$$

💰 **Revenue per Visit** - shows financial impact per visit from the button change

$$\text{RPV} = \frac{\text{Total Revenue}}{\text{Total Visits}}$$

4. Which metric should we choose to compare the different versions?



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